

Communication-Based Distributed Control of the Stacked Polyphase Bridges Converter

Lebing Jin¹, Staffan Norrga, *Member, IEEE*, Oskar Wallmark¹, *Member, IEEE*, and Nikolaos Apostolopoulos

Abstract—The stacked polyphase bridges converter has gained increasing interest in recent years due to its modular design and easy adaption to different power levels. The real-time control and dynamic data transmission of a stacked polyphase bridges converter consisting of a number of submodules are challenging tasks. For the reason that the distributed control architecture can maintain the benefit of modularity, such an approach dedicated for the stacked polyphase bridges converter is presented in this paper. Suitable communication techniques and synchronization techniques for the presented architecture are discussed. In addition, based on a dynamic model of the converter, the sensitiveness of the distributed control system to communication bandwidth and delay is studied on an experimental prototype.

Index Terms—Distributed control, modular converter, serial peripheral interface, stacked polyphase bridges (SPB) converter.

I. INTRODUCTION

DURING recent years, modular converters have been attracting increasing interest in industry [1]. Due to its simplicity, the traditional centralized control has been adopted for modular converters in many industrial applications as well as in academic research. However, since the central controller executes the whole control algorithm, high CPU computation speed and high communication bandwidth are usually required [2], [3]. Besides, the modularity of the modular converter can be limited by the centralized control [4]. In order to overcome these drawbacks, distributed control architectures are being pursued for modular converters. Much of the literature investigating such an approach has focused on the modular multilevel converter (MMC) for high-voltage applications, such as high-voltage direct current transmission [4]–[7], wind power [8], and photovoltaics [9]. In these applications, considering the large number of submodules and high potential difference between

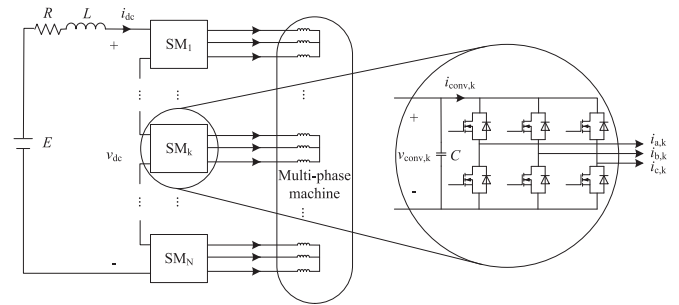


Fig. 1. Structure of the SPB converter topology.

the submodules, optical fiber and high-speed communication technology are usually adopted. Another focus is on parallel-connected converters, such as ac/dc converters [10]–[14]. To improve the performance of the distributed control system for modular converters, research has also been conducted on fault-tolerant designs [7], [8], [13], timing constraints analysis [6], and analysis of noncharacteristic harmonic components owing to synchronization error [15].

The stacked polyphase bridges (SPB) converter (see Fig. 1) is a type of modular topology that was introduced recently. The SPB converter has been proposed for use in applications, such as electric vehicles [16], [17], transformerless offshore wind turbine [18], [19], and potentially also as compact drives for pumps, fans, and compressors. This topology has many inherent benefits, with one particular advantage being its modular structure, which enables system scalability to be easily implemented through a series connection of different numbers of submodules, and potentially provides fault tolerance.

To achieve a truly modular implementation of the SPB converter, it is important that each submodule is designed as a standalone unit, which is equipped with the necessary dc-link capacitor, power bridges with gate drivers, and signal circuitry. Moreover, there are considerable benefits in equipping each submodule with a microcontroller so that the pulse width modulation (PWM) pulse generation and required control algorithm can be distributed among the series-connected submodules. The submodules can preferably be linked together through a data network. The converter can, thus, be implemented in a new application without doing major design modifications. In [19]–[21], theoretical control algorithms (particularly the dc-link control) for SPB converters are discussed. In [17] and [22], it is demonstrated that phase-shifted carrier-based modulation is beneficial

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L. Jin, S. Norrga, and O. Wallmark are with the Department of Electric Power and Energy System, Electrical Engineering, KTH Royal Institute of Technology, Stockholm 11428, Sweden (e-mail: lebingj@kth.se; norrga@kth.se; owa@kth.se).

N. Apostolopoulos is with Maclaren, London SW1X 7LJ, U.K. (e-mail: nkapostolopoulos@gmail.com).

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for minimizing the harmonics in the dc-link current, hence reducing the size of the dc-link capacitor. Despite these research efforts, the strategy on the realization of the distributed control architecture in practice and, more importantly, the study on the sensitiveness of the control scheme to the data network limitations have so far been lacking. The aim of this paper is to fill that void. This paper presents a general communication-based distributed control architecture for a modular inverter system. The communication sequence and delays in such a distributed architecture are analyzed. Moreover, it also shows how the stability impact of a communication delay in the controller can be analyzed using the Nyquist criterion. The discussions in this paper can be easily adapted to other modular converter systems.

The outline of this paper is as follows. Section II describes the distributed control architecture of the SPB converter and discusses the communication algorithm and synchronization issues. Particularly, the impact of the communication bandwidth and delay is also analyzed in this section. Experimental results are presented and discussed in Section III. Finally, a conclusion is given in Section IV.

II. ARCHITECTURE OF THE DISTRIBUTED CONTROL SYSTEM

A. Basics of the Operation of the SPB Converter

The SPB converter is depicted in Fig. 1. It consists of N series-connected submodules, each comprising a conventional two-level, three-phase converter equipped with low-voltage semiconductor. The ac terminals are connected to a multistar load, such as a multiphase machine. From Fig. 1, the current dynamic is derived as

$$L \frac{di_{dc}}{dt} = E - Ri_{dc} - \sum_{k=1}^N v_{conv,k} \quad (1)$$

where E , L , and R are voltage, inductance, and resistance of the voltage source, respectively, i_{dc} is the dc-link current, and $v_{conv,k}$ is the capacitor voltage of submodule k . It is assumed that all submodule capacitors have the same capacitance C . Thus, for the k th submodule, the capacitor-voltage dynamic is derived as

$$C \frac{dv_{conv,k}}{dt} = i_{dc} - i_{conv,k} \quad (2)$$

where $i_{conv,k}$ represents the input current of the k th submodule.

B. Control Algorithm

As each submodule of the SPB converter is a two-level, three-phase inverter, a standard field-oriented current control is adopted. Besides, a dc-link voltage control is required to balance the capacitor voltage of each submodule. Fig. 2 depicts the control algorithm for the SPB converter. A brief overview of the most important control blocks is given in the following.

The field-oriented current control [23] is implemented separately for each submodule converter. As illustrated, it includes three parts: proportional-integral (PI) controller, dq decoupling,

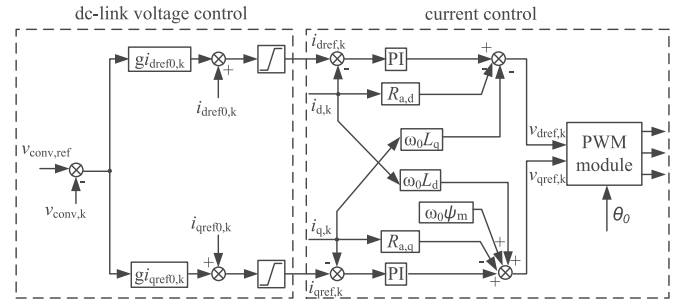


Fig. 2. Control algorithm.

and active damping. The dq voltage references are generated as

$$\mathbf{v}_{dq,k}^{\text{ref}}(s) = \mathbf{F}(s)(\mathbf{i}_{dq,k}^{\text{ref}}(s) - \mathbf{i}_{dq,k}(s)) + \mathbf{W}(s)\mathbf{i}_{dq,k}(s) \quad (3)$$

$$\mathbf{F} = \begin{bmatrix} k_{p,d} + k_{i,d}/s & 0 \\ 0 & k_{p,q} + k_{i,q}/s \end{bmatrix} \quad (4)$$

$$\mathbf{W} = \begin{bmatrix} -R_{a,d} - \omega_0 L_q \\ \omega_0 L_d - R_{a,q} \end{bmatrix} \quad (5)$$

where k_p , k_i , and R_a are control parameters; their tuning follows the criteria in [23].

The principle of the dc-link voltage control is to adjust the power independently for each submodule to control the capacitor voltage deviation from the reference value, i.e., v_{conv}^{ref} .

$$\mathbf{i}_{dq,k}^{\text{ref}} = \mathbf{i}_{dq,k}^{\text{ref0}} + g\mathbf{i}_{dq,k}^{\text{ref0}}(v_{conv}^{\text{ref}} - v_{conv,k}) \quad (6)$$

where $\mathbf{i}_{dq,k}^{\text{ref0}}$ are set by the conventional flux and torque controllers; they are common to all the submodules. Here, g is the stabilization gain.

In normal operation, balanced submodule capacitor voltages are most beneficial for equalization of stress on circuit components. Therefore, v_{conv}^{ref} can be chosen as v_{dc}/N .

To summarize, the control structure for each submodule can be divided into two levels as follows.

- 1) The outer voltage control balances the submodule active power.
- 2) The inner current control controls active and reactive power in a decoupled way.

Remark: Modular permanent magnet synchronous motor (PMSM) design suitable for the SPB converter topology is considered in this paper [24]. The PMSM model in Fig. 2 can account also for a synchronous reluctance machine by letting $\psi_m = 0$. With ψ_m as the rotor-flux linkage, it can account for a rotor-flux-oriented induction machine by letting $L_d = L_q = L_\sigma$, where L_σ is the total leakage inductance. By letting $\omega_0 \psi_m = 0$, it accounts for an RL load.

C. Distributed Control Architecture

Based on the above control algorithm, a distributed control architecture consisting of three parts, namely: 1) master controller, 2) slave controllers, and 3) data network, is proposed and presented in Fig. 3. Each submodule can be divided into two parts: signal module and power module. The microcontroller, gate-drive circuit, and measurement circuit are integrated in the

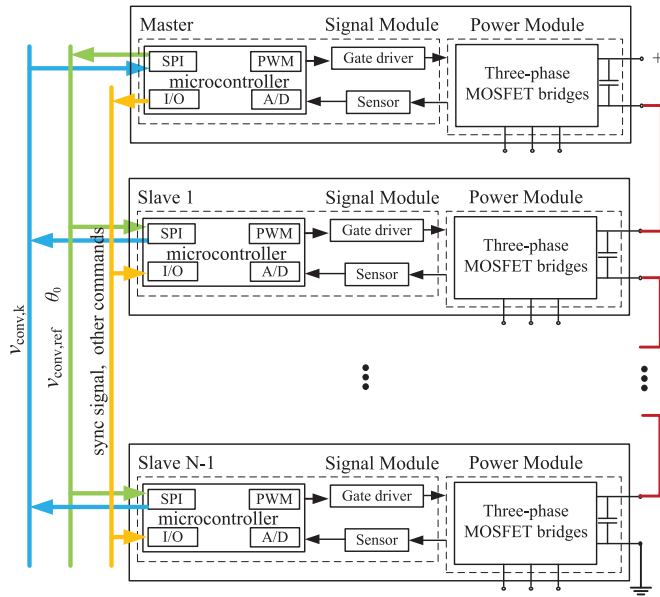


Fig. 3. Schematic of an N -submodule SPB.

signal module. Low-voltage MOSFET-based three-phase bridges with a small dc-link capacitor make up the power module. The communication bus and signal I/Os are galvanically isolated between the submodules.

Slave controllers are responsible for calculating the PWM intervals and creating the associated gate drive pulses. They can be implemented using microcontrollers with enough computation capability of doing relatively simple programming. It is also desirable to have the slave controllers programmed to send dc-link capacitor voltage samples and fault messages back to the master controller for system regulation and monitoring. In general, slave controllers execute the following submodule-level tasks:

- 1) measure the capacitor voltage and phase currents;
- 2) detect phase-leg faults, for instance, over-current;
- 3) regulate dq currents;
- 4) generate PWM signals.

Besides the submodule-level control, the functions of the master controller are to send the common voltage reference and PWM synchronization signal. It can be implemented using the same microcontroller or using a more advanced microcontroller for high-demand applications, requiring sophisticated real-time control and system condition monitoring. The master device in this paper deals with the following converter-level tasks:

- 1) generate the capacitor voltage reference;
- 2) balance submodule voltages;
- 3) obtain rotor position signal;
- 4) transmit the voltage reference and the rotor position to slaves;
- 5) send out other commands, such as the PWM synchronization signal (which will be further discussed in Section II-E).

Compared to the centralized control architecture, PWM signals in the distributed system are generated locally in the submodules, thus highly reducing the bandwidth requirements for

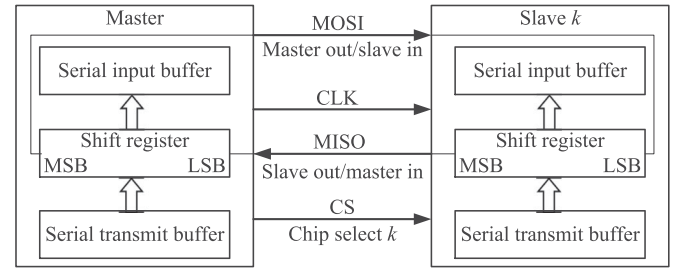


Fig. 4. SPI master-slave connection.

the communication. In addition, voltage balancing and current control can be done locally since the voltage and the currents are measured locally, reducing in this way communication load and delay.

D. Communication Technique

Digital communication is widely used in industrial applications. Several communication protocols have been adapted for distributed control of modular converters, such as the Ethernet for control automation technology for MMCs with fairly large numbers of submodules [6], [7], the controller area network (CAN bus) for automotive applications with relatively low bandwidth requirements [4], and the serial peripheral interface (SPI) for transferring data between multiple processors [25].

The SPI is a high-speed, single-master, synchronous communication interface and it is used in this paper for implementing the data network. Fig. 4 illustrates an SPI connection between the master and a slave. The SPI usually allows a serial bit stream of preprogrammed length (up to 16 bits) to be transmitted to the slaves at a programmed baud rate (up to 10 Mb/s).

As seen in Fig. 3, the master controller and $N - 1$ slave controllers are linked together by the data network, i.e., the SPI communication in this paper. According to the aforementioned distributed control architecture, the inputs for the inner current control are all measured locally and do not require communication, whereas the PWM and the voltage balance control need a voltage reference, which is calculated in the master device from the capacitor voltages of all submodules, thus communication is needed. Besides, the rotor position angle θ_0 could also be transmitted via the data network. In this way, the resolver signal is only fed into the master device reducing the hardware connections. An example communication sequence is exemplified in Fig. 5 in which four submodules are considered. Note that only the cyclic payload is considered. The control variables are transmitted to the slaves via the SPI bus in the serial mechanism. Letting one message represent one transmitted variable, for a four-submodule SPB converter, six messages are needed during one CP. In this paper, each message is 64 bits, consisting of 32 bits of data, 16 bits of error information, and 16 reserved bits.

E. Submodule Carrier Synchronization

In the distributed control system, all submodules are linked through a data network with communication protocols designed

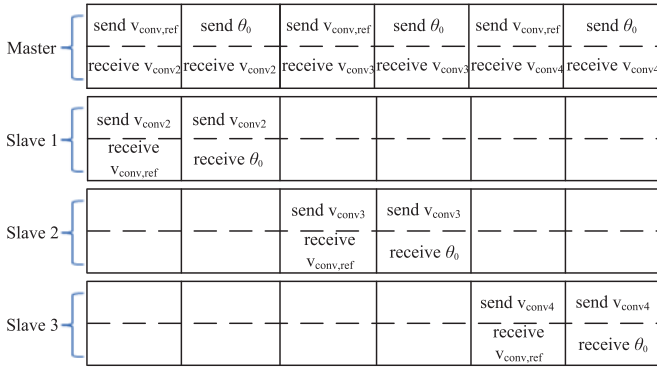


Fig. 5. Example communication sequence for a four-submodule SPB converter.

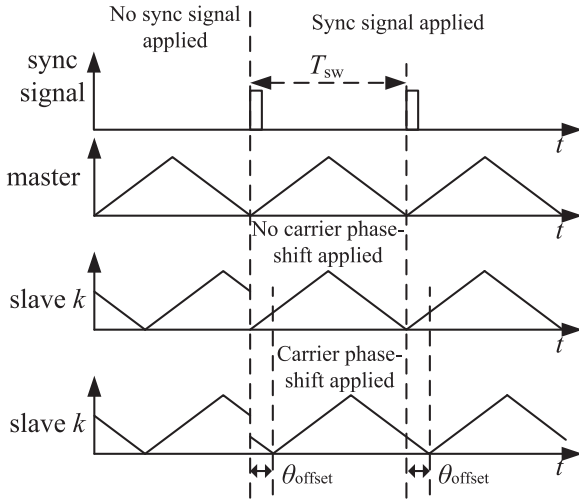


Fig. 6. Carrier synchronization and interleaving between different PWM modules.

for real-time control and synchronization of the submodules. Carrier synchronization is particularly important for carrier-based modulation in cascaded modular converters since harmonic cancelation in the dc-side current can only be achieved with proper synchronization, thus minimizing the dc-link capacitance [15], [22], [25].

Take the digital signal processor (DSP) as an example. The general-purpose timers embedded in the DSP can be programmed to implement the sampled triangular carriers needed for PWM generation. In order to cancel certain high-order harmonics as reported in [22], the three carriers for each submodule should be in phase, while the carriers should be phase-shifted $2\pi/N$ rad among submodules. The processor card in use has a crystal oscillator as a clock generator. Although the oscillator has a very small manufacturing tolerance, the frequency error accumulates so fast that a phase delay among the PWM carriers of the submodules cannot be ignored. The most common way to maintain the synchronization of the modular system is that the master controller is arranged to send a synchronization signal every switching cycle T_{sw} (see Fig. 6) to slave devices through the signal network. All the slave processors measure and automatically compensate their phase misalignments. For

interleaving modulation, a desirable θ_{offset} can be implemented at this moment. This synchronization signal is fairly easy to implement, as seen in Fig. 3 using a digital I/O pin on the master device.

F. Real Time and Stability Analyses

Supposing that the master controls $N - 1$ slaves and that the communication sequence in Fig. 5 is employed, the predefined communication period (CP) in master device (T_{mcp}) and slave device (T_{scp}) can be expressed as

$$T_{mcp} = M(N - 1)T_{cs} \quad (7)$$

$$T_{scp} = MT_{cs} \quad (8)$$

where M is the number of variables that needs to be transferred ($M = 2$ in this paper), and T_{cs} is the communication time slice for one variable in Fig. 5.

In Fig. 7, the time sequence between the master and $N - 1$ slaves is illustrated. The transmission of a variable in Fig. 5 is marked as a single message here. Generally, the communication time slice T_{cs} (further illustrated in Fig. 8) includes the delay time before the valid data are setup, the execution time of the data transfer, the propagation time, and the gap phase (waiting time) between the transmissions of two bytes [26]:

$$T_{cs} = T_M + T_S + 4 \times T_{d.pg} + T_{d.wait} \quad (9)$$

$$T_M = T_{d.MO} + T_{exec}/2 \quad (10)$$

$$T_S = T_{d.SO} + T_{exec}/2 \quad (11)$$

where time variables are defined as follows:

- 1) T_M —total required time for a single transmission arising from the master;
- 2) T_S —total required time for a single transmission arising from the slave;
- 3) $T_{d.pg}$ —propagation delay arising from the isolator (in the case isolators are implemented) and the cable (note that this is neglected in this paper because the communication cable is very short).
- 4) $T_{d.wait}$ —waiting time after one transmission;
- 5) $T_{d.MO}$, $T_{d.SO}$ —delay time before valid data are sent out from either the master or the slave. Only exists at the transmitter side;
- 6) T_{exec} —execution time of data transfer. Proportional to the baud rate of the SPI communication.

Obviously, due to the nature of the serial communication, communication delay, i.e., T_d in Fig. 7, is inevitable and may influence the dc-link stability. Take slave 1 as an example. The locally measured capacitor voltage $v_{conv,1}$ gets updated in slave 1 and is transmitted to the master at t_1 . However, the corresponding updated voltage reference v_{conv}^{ref} for the dc voltage control will not be received in slave 1 until next CP. Figs. 5 and 7 indicate that this time delay between the locally measured $v_{conv,k}$ and the received reference v_{conv}^{ref} in the slave devices can be expressed as

$$T_d = T_{mcp}. \quad (12)$$

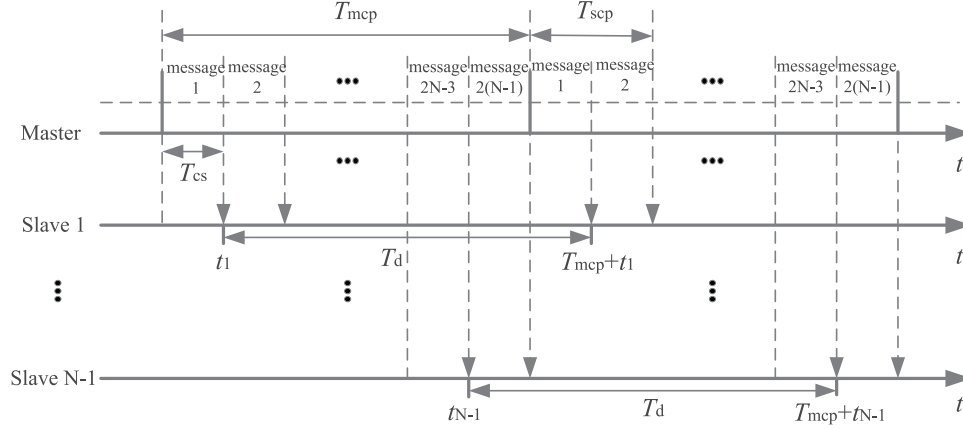


Fig. 7. Time sequence between the master and slaves.

This delay gets longer as the number of transmitted variables increases, as also the baud rate decreases since T_{exec} in (10) and (11) increases.

Furthermore, the voltage reference fed into the controller is actually sampled data, owing to the fact that the received control variables, i.e., v_{conv}^{ref} and θ_0 , in slaves get updated every T_s . Thus,

$$v_{conv}^{ref'}(t) = \sum_{n=0}^{+\infty} v_{conv}^{ref}(nT_s) \delta(t - nT_s) \quad (13)$$

$$= v_{conv}^{ref}(t) \sum_{n=0}^{+\infty} \delta(t - nT_s) \quad (14)$$

where $v_{conv}^{ref'}$ is the sampled voltage reference, T_s is the sampling time, and ideally equal to T_{mcp} in this case.

In [21], the following linearized converter model is presented to analyze the dc-link stability (note that v_{conv}^{ref} is chosen to be v_{dc}/N in this model):

$$L \frac{d\Delta i_{dc}}{dt} = -R\Delta i_{dc} - \Delta v_{dc} \quad (15)$$

$$C \frac{d\Delta v_{dc}}{dt} = N\Delta i_{dc} + \frac{P^* \Delta v_{dc}}{v_{conv}^*{}^2} - \frac{g' \Delta v_{dc}}{v_{conv}^*} \quad (16)$$

where the dc controller gain $g' \approx P^*/v_{conv}^*$. The Δ notation represents deviations around some operating point, (e.g., $\Delta x = x - x^*$, where x^* is the operating point). Assuming all submodules operate with the same active power, we get $P_k = P^* + \Delta P_k$, $v_{conv,k} = v_{conv}^* + \Delta v_{conv,k}$. P_k is the power into the k th submodule.

The influence of a low-pass filter on the above model has been taken into account in [21], suggesting that the resulting stability can be analyzed by applying the Nyquist criterion to the system. Similar to that, considering the aforementioned two issues in the dc-link control, i.e., the communication delay and bandwidth in Fig. 9, (16) can be expressed as

$$C \frac{d\Delta v_{dc}}{dt} = N\Delta i_{dc} + \frac{P^* \Delta v_{dc}}{v_{conv}^*{}^2} - \frac{g'}{v_{conv}^*} [1 - D(s)B(s)] \Delta v_{dc} \quad (17)$$

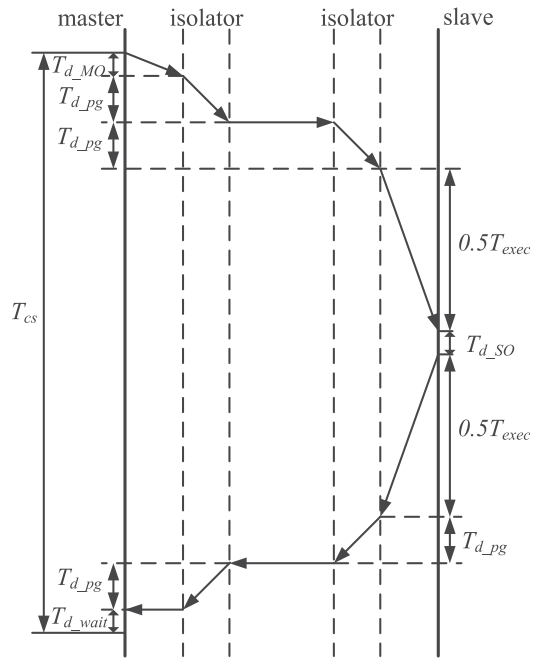


Fig. 8. Structure of the communication time slice.

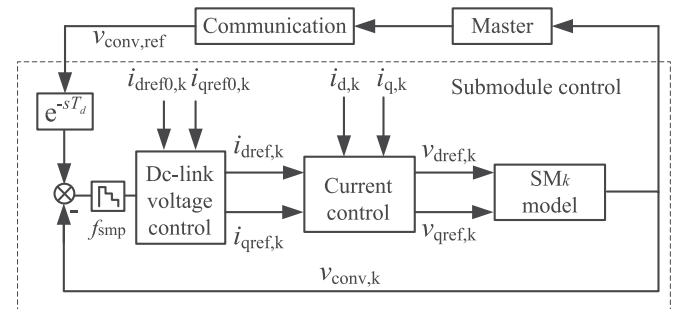


Fig. 9. Submodule control with communication bandwidth and delay.

where $D(s) = e^{-sT_d}$ represents a time delay of T_d seconds due to serial communication and $B(s) = \frac{1-e^{-sT_s}}{s}$ is the transfer function used to model the sampled control variables from communication [27], [28].



Fig. 10. Experimental setup: Four-submodule SPB converter.

Equivalently, (15) and (17) can be expressed as a negative-feedback system

$$\Delta v_{dc} = -G(s)\Delta v_{dc} \quad (18)$$

where

$$G(s) = \frac{1}{sC} \left\{ \frac{1}{sL + R} - \frac{P^*}{v_{conv}^{*2}} + \frac{g'}{v_{conv}^*} \left[1 - \frac{e^{-sT_d}(1 - e^{-sT_s})}{s} \right] \right\}. \quad (19)$$

The resulting stability of the distributed control system can thus be analyzed by applying the Nyquist criterion to (19). If the Nyquist curve does not encircle -1 , system stability can be maintained.

III. IMPLEMENTATION AND EXPERIMENTAL VERIFICATION

Experiments have been implemented to evaluate the performance of the distributed control system. In the setup (see Fig. 10), four submodules are connected to four sets of three-phase passive loads. Each submodule is integrated on a single printed circuit board (PCB). Each PCB is equipped with a DSP TMS320F28069 and the internal SPI module is adopted as the communication interface. The communication bus is isolated by ISO7241 digital isolators among submodules. Silicon MOSFETs are used as switches with 20 kHz switching frequency. Two-phase currents and the dc capacitor voltage are locally measured on each submodule and the measurements are fed into the analog-to-digital converter in the DSP.

All the submodules are designed identical, whereby they can operate as either master or slave. In the software, one submodule is configured as master, while the other three are configured as slaves. The main parameters of the setup are listed in Table I.

A. PWM Synchronization and Interleaving

Fig. 11(b) presents the measured PWM signals of three submodules for the case that the synchronization technique is applied. As can be seen, the PWM synchronization between submodules is successfully achieved. Moreover, Fig. 11(c) presents

TABLE I
PARAMETERS OF THE SETUP UNDER TEST

Parameter	Value	Unit	Symbol
DC power supply	200	V	E
Number of submodules	4	—	N
Submodule capacitor voltage	50	V	$v_{conv,k}$
Tested ac peak current	6	A	I_0
Submodule dc capacitance	94	μ F	C
Switching frequency	20	kHz	f_{sw}
Fundamental frequency	50	Hz	f_0
Load resistance	3.2	Ω	R_{ac}
Load inductance	4.67	mH	L_{ac}

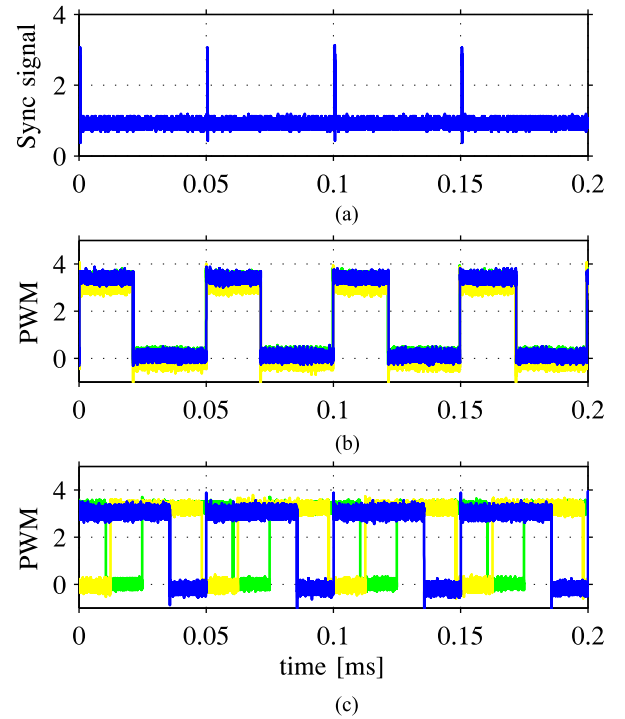


Fig. 11. Synchronization technique. (a) Synchronization signal. (b) PWM outputs are synchronized. (c) PWM outputs when carriers are interleaved by $\pi/2$.

the measured PWM signals of three submodules for the case that the interleaving technique is applied. As seen, the desirable phase shift is obtained.

B. Timing Constraints

To understand the timing constraints of the developed distributed system, an experimental test with two submodules is designed: one works as the master while the other as the slave. For evaluating the physical transmission time slice, i.e., T_{cs} , between two submodules, the communication scheme in Fig. 7 was adopted. In the experimental prototype, ribbon cables up to 10 cm in length were used as the communication connection between two submodules. The propagation delay from the cable can be neglected in this test, while the propagation delay from the digital isolators is maximum 42 ns according to the datasheet. The waiting time between transmissions of two 16-bit streams was programmed to be 10 SPI clock periods in

TABLE II
COMPARISON BETWEEN THE ESTIMATED AND THE MEASURED
COMMUNICATION TIME SLICE

Baud Rate (Mb/s)	Estimated T_{cs} (μs)	Measured T_{cs} (μs)
2.86	42.59	49.4
1.54	93.78	101.6
0.85	122.6	131.3

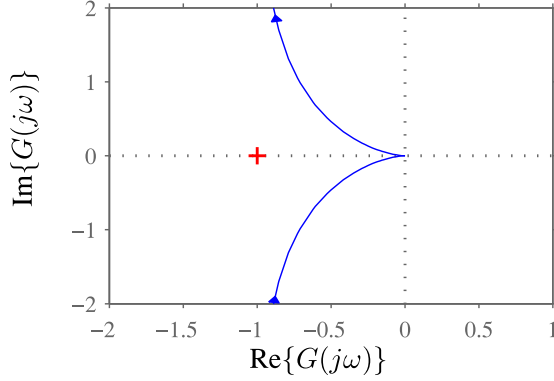


Fig. 12. Nyquist diagram for $G(j\omega)$.

this paper. Once the message was sent to the SPI register in the master device, a general purpose I/O pin would immediately be set to be low and back to high again when the master device completely received the message from the slave. The low period of the I/O signal can thus represent the communication time slice of one message. The measured results with different baud rates are listed in **Table II**, compared with the estimated T_{cs} from (9). The correctness of the timing analysis can thus be validated. The interrupt mechanism in the DSP creates delays when measuring T_{cs} from an I/O pin. Therefore, the measured values are a little larger than the calculated ones.

Consequently, the time delay can be easily estimated by (7) and (12). As an example, the whole system with four submodules is considered. The communication bandwidth is 2.86 Mb/s. The theoretical time delay can be obtained as follows:

$$T_d = 2 \times (4 - 1) \times T_{cs} = 255.54 \mu s. \quad (20)$$

The Nyquist criterion is applied to this system. The dc-side parameters are $L=1 \mu H$ and $R=0.1 \Omega$. In addition, $v_{conv}^* = 50 V$, $P^* = 250 W$, and $g' \approx 5$. The Nyquist curve in **Fig. 12** is obtained for $T_d=255.54 \mu s$. As seen, the Nyquist curve does not encircle -1 , so system stability is maintained. This provides a useful way to check the dc-link stability of the distributed system at any baud rates. An interesting conclusion is that the system stability is still maintained when the communication bandwidth is lower than 1 Mb/s, indicating that the CAN bus could be adopted. This would be beneficial owing to its simpler hardware connection and higher noise immunity.

C. Control Performance

1) Transient Responses of the System: Rapid torque response is generally essential to an automotive drive system.

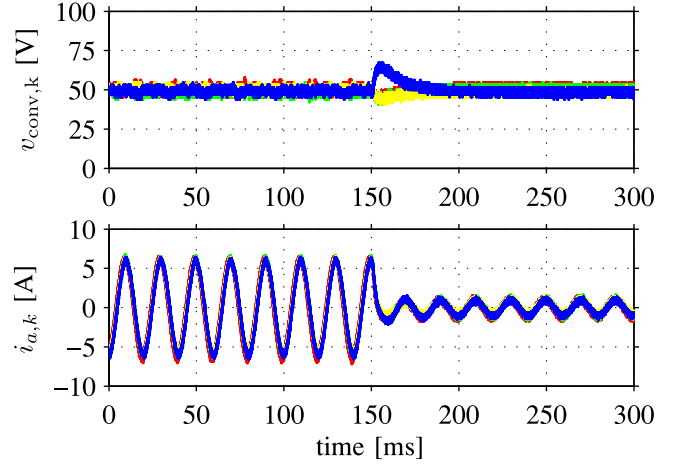


Fig. 13. Sudden reduction in the torque reference. Red: submodule 1; green: submodule 2; yellow: submodule 3; blue: submodule 4.

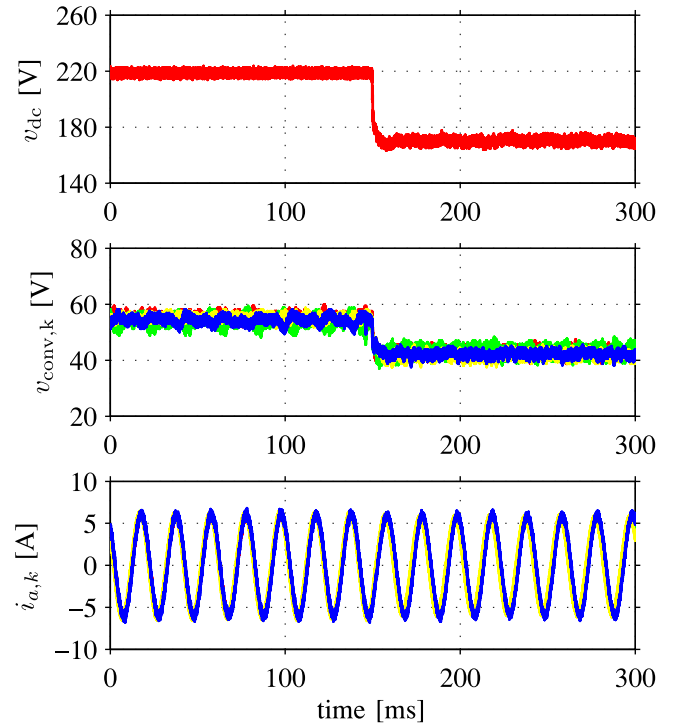


Fig. 14. Sudden drop in the dc bus voltage. Subplot 1: v_{dc} ; Subplot 2 and 3: $v_{conv,k}$ and $i_{a,k}$; red: submodule 1; green: submodule 2; yellow: submodule 3; blue: submodule 4.

Here, this is emulated by applying a sudden step in the q -axis current reference. **Fig. 13** shows the transient response of the system by changing the i_{qref} from 6 to 1 A. The waveforms of the submodule capacitor voltages and the ac load currents are depicted in the figure. It can be seen that an imbalance, caused by the communication delay in the submodule voltages, occurs at the beginning of the transient but is eliminated rapidly owing to the system control.

Fig. 14 shows the effects of a severe disturbance on the dc bus voltage. The waveforms of the dc bus voltage, submodule

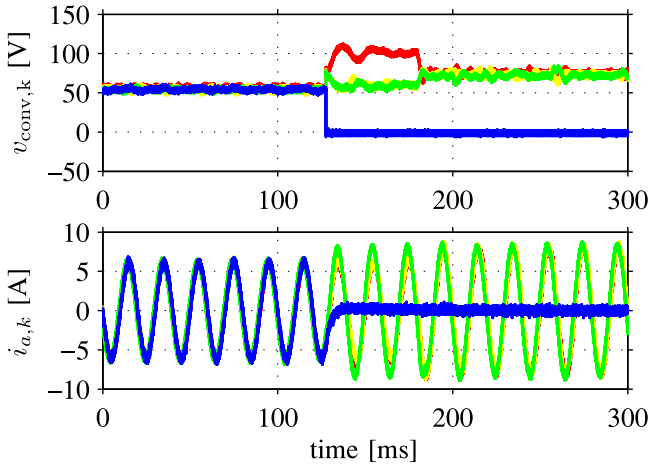


Fig. 15. Submodule four assumed failed and bypassed while maintaining total power constant in post-fault operation. Red: submodule 1; yellow: submodule 2; green: submodule 3; blue: submodule 4.

voltages, and ac load currents are illustrated in the figure. As seen, due to the dc voltage drop (of about 23%), $v_{conv,k}$ are adjusted by the system control within a few milliseconds with virtually no influence on the ac side.

2) Redundant Operation of the System: Normally, there are four submodules in the system to share the dc bus voltage. In case of a single submodule fault, the remaining healthy submodules have to share the dc bus voltage and the total power. In this situation, the voltage rise generally does not destroy the semiconductor devices since they are usually rated with a 50%–100% margin. The response is shown in Fig. 15 in which submodule 4, i.e., slave 3, is assumed to be the faulty one and is bypassed. After a 50 ms transient, the remaining submodules share the dc bus voltage and the system enters a new steady state. Thus, the system can still operate to supply the output power demand under a single submodule fault condition. In this case, submodule 1 is configured as the master controller. The master detects system errors and send out the rebalancing command. The 50 ms delay is caused by three reasons. First, it is a limitation of serial communication. The voltage reference gets updated every T_{mcp} . The more the messages transmitting among submodules, the longer this delay is. Second, a relatively long waiting time (a few tens of SPI CLK) is currently implemented between two adjacent transmissions to ensure the communication quality. Third, the data are in 32-bits floating format in the employed processor. All the 32-bits information (integer part + decimal part) are transmitted in the current communication protocol. The communication cycle can be reduced if the decimal parts of the variables are neglected in the transmission, in this way the delay can be reduced.

3) Motor Load: Experimental results of the control and system working with a motor load are shown in this section. The rotor position was measured by a resolver and fed into the master board. The employed four-pole PMSynRel machine was designed to allow for different connections of the 12 available windings. Therefore, each of the four submodules is able to control a set of independent three-phase windings. More

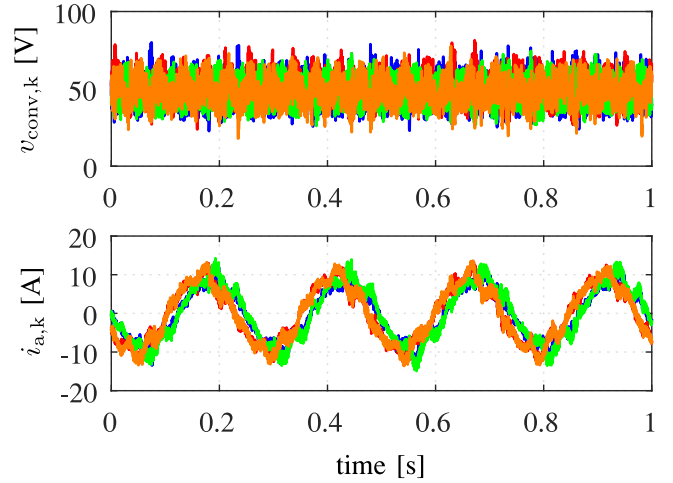


Fig. 16. Steady-state operation for a four-submodule setup driving a PMSynRel motor at 4 Hz electrical speed, 200 V DC bus voltage. Top: submodule capacitor voltages; bottom: phase current for each submodule. Blue: submodule 1; red: submodule 2; green: submodule 3; yellow: submodule 4.

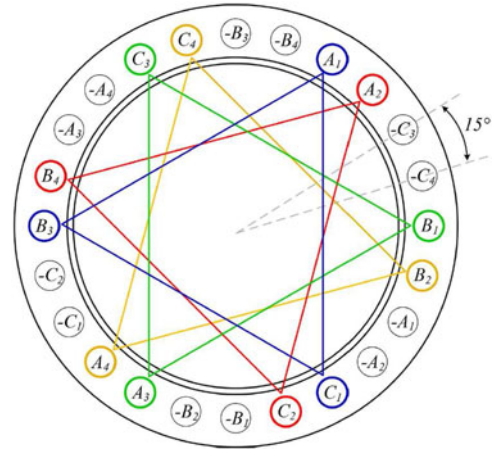


Fig. 17. Winding arrangement of the PMSynRel machine.

information on the motor design and characteristics can be found in [29]. Fig. 16 shows the submodule capacitor voltages and phase currents for the four SPB submodules during steady-state operation. The current waveforms for the four boards have 30° electrical phase shift due to the symmetrical winding configuration, as shown in Fig. 17. The four submodules are connected so that submodule connected to A_1 and the submodule connected to A_3 are in phase and the submodules connected to A_2 and A_4 are in phase. However, module A_1 and A_3 are phase shifted 15 mechanical degrees (i.e., 30 electrical degrees) with respect to A_2 and A_4 . This causes the relatively large oscillation in the submodule capacitor voltages.

IV. CONCLUSION

In this paper, an SPI-based distributed control system for a modular inverter system was developed. This control scheme was successfully implemented in a four-submodule prototype. The experimental results demonstrated the carrier synchronization technique and the dynamic behavior of the

presented distributed control system. In addition, time delays arising from the data transmission were theoretically analyzed and experimentally tested. The good match between them verifies the timing analysis. By taking into account the effects of communication delay and bandwidth, the dc-link stability of the SPB converter was analyzed and experimentally proven to be maintained even when the communication baud rate was lower than 1 Mb/s. This indicates that the CAN bus can also be adopted as the communication interface for this topology. The analysis in this paper provides a feasible way to choose the required communication technique for the distributed control of modular converters.

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Lebing Jin was born in Shaanxi, China, in 1988. She received the B.Sc. degree in electrical engineering from Tianjin University, Tianjin, China, in 2010, the M.Sc. degree in electrical engineering from Tsinghua University, Beijing, China, in 2012, and the Ph.D. degree in electrical engineering from the KTH Royal Institute of Technology, Stockholm, Sweden, in 2016.

She is currently with the telecommunication industry, Stockholm, Sweden. Her research interests include power electronic converters, harmonic analysis, and control of electric drives.



Staffan Norrga (S'00–M'06) was born in Lidköping, Sweden, in 1968. He received the M.Sc. degree in applied physics from Linköping Institute of Technology, Linköping, Sweden, in 1993, and the Ph.D. degree in electrical engineering from the KTH Royal Institute of Technology (KTH), Stockholm, Sweden, in 2005.

Between 1994 and 2011, he was a Development Engineer at ABB, Västerås, Sweden, in various power-electronics-related areas, such as railway traction systems and converters for HVdc power transmission systems. He is currently an Associate Professor at KTH. He is the inventor or coinventor of 12 granted patents and 13 patents pending, and has authored or coauthored more than 75 scientific papers published at international conferences or in journals. His research interests include new converter topologies for power transmission applications and grid integration of renewable energy sources.



Oskar Wallmark (S'01–M'06) received the Ph.D. degree in electric power engineering from Chalmers University of Technology, Göteborg, Sweden, in 2006.

He is currently an Associate Professor in the Department of Electric Power and Energy Systems, KTH Royal Institute of Technology, Stockholm, Sweden. His research interests include control and analysis of electric drives in electric and hybrid electric drive trains.



Nikolaos Apostolopoulos was born in Greece in 1988. He received the Diploma degree in electrical and computer engineering from the National Technical University of Athens, Athens, Greece, in 2012, and the M.Sc. degree in electric power engineering from the KTH Royal Institute of Technology, Stockholm, Sweden, in 2015.

He is currently with the motorsport industry, London, U.K. His current research interests include control of high-performance electric drives for automotive applications, modular converters,

and power electronics.